Assignment 3 -
Software Measurement III- Function Point

Group 2

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Step 1: Link GQIM Goals to FP Metrics

Goal 1: Transaction Accuracy

- *Question:* Are all transactions processed and reconciled correctly?
- *Indicator:* Accuracy Ratio
- *Metric (FP Link):* Count EI (Create or Update Transaction), EO (Account Statement), and ILF (Transaction Records)
- *DET / RET / FTR:* Transaction fields, account summary tables, and linked ledgers
- *Result:* FP measurement reflects the functional effort required to achieve accurate and reconciled transactions

Goal 2: Data Integrity

- *Question:* How consistent and traceable are database changes?
- *Indicator:* Integrity Score
- *Metric (FP Link):* Count ILFs (Audit Log), EIFs (Backup Repository), and EOs (Data Integrity Reports)
- *DET / RET / FTR:* Log identifiers, change types, authorization flags, and checksum fields that reference ILFs and EIFs.
- *Result:* Function points represent the logical data files and outputs that ensure traceability and audit completeness.

Goal 3: Access Security

- *Question:* How effectively are authentication and authorization controls enforced?
- *Indicator:* Security Index
- *Metric (FP Link):* Count EI (Login & Logout), EQ (Access Status Query), ILF (User Access Events), and EIF (External Identity Provider)
- *DET / RET / FTR:* User identifiers, MFA tokens, encryption states, and role records that are referenced by FTRs.
- *Result:* FP counts reflect the security functions implemented through input, inquiry, and external identity interactions.

Goal 4: System Availability

- *Question:* How reliable and responsive is the platform under load?
- *Indicator:* Availability Rate
- *Metric (FP Link):* Count EO (Availability Reports), EQ (Status Dashboards), EIF (Monitoring Service API)
- *DET / RET / FTR:* Uptime data, incident identifiers, and service type records that are referenced from monitoring tools.
- *Result:* FP elements capture the system outputs and queries that enable monitoring and uptime analysis.

Goal 5: Compliance and Audit Readiness

- *Question:* How well does the system adhere to regulatory controls?
- *Indicator:* Compliance Score
- *Metric (FP Link):* Count EO (Audit Reports), EQ (Control Queries), and ILF (Compliance Control Table and Audit Result Logs)
- *DET / RET / FTR:* Control identifiers, test results, severity ratings, and compliance records.
- *Result:* FB counts quantify the internal files and outputs used to demonstrate regulatory compliance and audit readiness.

Step 2: Identify Function Types and Assess Complexity

1) Identify and List ILFs, EIFs, EIs, EOs, EQs:

- Transaction Accuracy
  - ILF – Transaction Records: Internal file storing all transaction details and reconciliation data
  - EI – Create/Update Transaction: External input capturing user or system-initiated transaction entries and updates
  - EO – Account Statement: Output summarizing processed transactions and balances for review or reconciliation.

- Data Integrity
  - ILF – Audit Log: Internal file capturing every database change event for traceability
  - EIF – Backup Repository: External file maintained outside the system that stores backup or replicated data
  - EO – Data Integrity Reports: Outputs report showing inconsistencies, checksums, or validation results.
- Access Security
  - ILF – User Access Events: Internal record of logins, logouts, and permission changes.
  - EIF – External Identity Provider: External authentication service (SSO)
  - EI – Login & Logout: External inputs capturing authentication credentials and session termination
  - EQ – Access Status Query – Inquiry to verify current session or authorization level without updating data
- System Availability
  - EIF – Monitoring Service API: External interface providing uptime and performance metrics.
  - EO – Availability Reports: System-generated outputs showing uptime percentages, downtime events, and performance statistics.
  - EQ – Status Dashboards: User-triggered inquiries retrieving current health and status metrics from the monitoring API.
- Compliance and Audit Readiness
  - ILF – Compliance Control Table / Audit Result Logs: Internal files containing compliance checks, control mappings, and audit outcomes.
  - EO – Audit Reports: Output presenting compliance verification results and severity ratings.
  - EQ – Control Queries: Inquiry functions that retrieve specific compliance or audit control statuses.

2) Count DET, RET, FTR, Assign Complexity and FP weight:

Record Types (RET): User recognizable subgroups of data elements within an ILF or an EIF. It is best to look at logical groupings of data to help identify them.

Data Element Types (DET): A unique user-recognizable, non-recursive (non-repetitive) field containing dynamic information. If a DET is recursive then only the first occurrence of the DET is considered, not every occurrence.

File Type Referenced (FTR): File type referenced by a transaction. An FTR must be an Internal Logical File (ILF) or External Interface File (EIF).

A- Transaction Accuracy:

1. Data Function Types (ILF/EIF use DET & RET)

Type	Name	RET	DET	Complexity (RET×DET matrix)	FP Weight*
ILF	Transaction Records	3	28	Average (RET 2–5 & DET 20–50)	10

1.1. ILF- Transaction Records

- RET is a logical, user-recognizable subgroup inside the file.
  - · RET-1 Header (identity & context)
  - · RET-2 Monetary / Fees (amounts, currencies, fees, FX)
  - · RET-3 Reconciliation / Approvals (recon/approval/dispute)
- DETs are counted once per file (not per row) and must be user-recognizable on screens/reports or maintained.

RET-1 Header (10 DET)	**RET-2 Monetary / Fees (9 DET)**	**RET-3 Reconciliation / Approvals (9 DET)**
transaction_id	amount	recon_status
acct_from	currency	recon_batch_id
acct_to	dr_cr_indicator	recon_date
txn_type	fee_amount	approval_required
channel	fee_currency	approval_status
description	tax_amount	approved_by
reference_no	tax_currency	approved_at
initiated_by	fx_rate	dispute_flag
created_at	running_balance_after	dispute_id
posting_date		
		Total DETs= 28

- Result: RET = 3, DET = 28 for the Transaction Records ILF.
- Matrix check (Slide 17–18): RET=2–5 & DET=20–50

⇒ Average complexity

2. Transaction Function Types (EI/EO use DET & FTR)

Type	Name	FTR	DET	Complexity (FTR×DET matrix)	FP Weight*
EI	Create/Update Transaction	3	14	High (FTR ≥3 & DET 5–15)	6
EO	Account Statement	3	22	High (FTR 2–3 & DET ≥20)	7

1.2. EI- Create/Update Transaction

- FTR is an ILF/EIF that the transaction reads or maintains.
  - Transaction Records (ILF) — maintained (create/update)
  - Account (ILF) — read/updated (e.g., available balance, link to account)
  - Ledger (ILF) — read/updated (posting to GL)
- Result: FTR = 3

Count DETs: Use the actual input screen/JSON for this EI; count each distinct field once.

1. acct_from
2. acct_to
3. amount
4. currency
5. type
6. channel
7. timestamp
8. description
9. reference_no
10. fee_flag
11. schedule_date
12. mfa_token_id
13. device_id
14. initiated_by

- Result: DET = 14, FTR = 3 for the EI.
- Notes: Primary keys auto-generated *don't count* unless they are entered/ recognized by users on this EI (Reference: Week 9 rule of user-recognizability).

2.2. EO- Account Statement: A derived output (e.g., PDF/email) summarizing activity and balances for a period.

- FTRs used to produce the output → FTR = 3
  - · Transaction Records (ILF) - line items
  - · Account (ILF) - account master, opening/closing balances
  - · Ledger (ILF) - postings/interest/fees roll-ups
- Result: FTR = 3

Count DETs (unique data elements shown on the output)

DETs on outputs are counted once per report layout (not per row repeated).

1. acct_no
2. acct_name
3. period_start
4. period_end
5. opening_bal
6. closing_bal
7. txn_id
8. txn_date
9. posting_date
10. txn_type
11. description
12. merchant_or_counterparty
13. debit
14. credit
15. running_bal
16. currency
17. fx_rate
18. fee_total
19. interest_total
20. dispute_flag
21. statement_id
22. generated_at

- Result: DET = 22, FTR = 3 for the EO.

- UG weights: ILF (L/A/H = 7/10/15), EIF (5/7/10), EI (3/4/6), EO (4/5/7), EQ (3/4/6).

B- Data Integrity:

Data Integrity slice: It makes every DB change traceable and verifiable (audit trail and backups) and reports mismatches.

1. Data Function Types (ILF/EIF use DET & RET)

Type	Name	RET	DET	Complexity (RET×DET)	FP Weight
ILF	Audit Log	2	14	Low (RET 2–5, DET 1–19)	7
EIF	Backup Repository	1	10	Low (RET=1, DET 1–19)	5

1.1. ILF- Audit Log

- Identify RETs (logical subgroups)- RET = 2

· RET-1: Change Event
· RET-2: Security Context

- Count DETs (user-recognizable fields)

Count a field once per file. Technical fields that users never see don't count; if they appear in admin UIs or reports, they do count.

· RET-1 Change Event (8 DET): log_id, entity_type, entity_id, change_type, before_value, after_value, reason_code, checksum

· RET-2 Security Context (6 DET): changed_by, changed_at, ip, session_id, mfa_present, signature

- Result: RET=2, DET=14 (ILF).

- Matrix: ILF with RET 2–5 & DET 1–19

⇒ Low complexity ⇒ weight 7 FP.

1.2. EIF- Backup Repository: External logical file (outside your app) that stores snapshots/backups you reference

- RETs = RET = 1

- Single logical subgroup of "backup metadata."

- DETs → DET = 10 = backup_id, object_type, object_id, snapshot_time, retention, checksum, storage_tier, location, size, status

- Result: RET=1, DET=10 (EIF).
- Matrix: EIF with RET=1 & DET 1–19

⇒ Low complexity ⇒ weight 5 FP.

2. Transaction Function Types (EI/EO/EQ use DET & FTR)

Type	Name	FTR	DET	Complexity (FTR×DET)	FP Weight
EO	Data Integrity Reports	2	15	Average (FTR 2–3, DET 1–19)	5

Data Integrity subtotal (UFC)= 7+5+5 = 17 FP

2.2. EO- Data Integrity Reports: A derived output listing integrity checks (mismatch, orphan, duplicate, checksum status, etc)

- FTRs (files referenced) = FTR = 2

  · Audit Log (ILF)
  · Backup Repository (EIF)

- DETs (unique data elements on the report) -DET = 15

object_type, object_id, last_change, last_backup, delta_hash, mismatch_flag, missing_rows, orphan_rows, duplicate_keys, checksum_status, sample_record_id, detected_at, severity, owner_team, remediation_status

Repeated rows don't multiply DETs; you count each column/element once per report layout. Derived values do count if shown (e.g., delta_hash, counts).

- Result: FTR=2, DET=15 (EO).
- Matrix: EO with FTR 2–3 & DET 1–19

⇒ Average complexity ⇒ weight 5 FP

C- Access Security:

Authentication and authorization visibility (logins, MFA, role changes; quick status checks).

1. Data Function Types (ILF/EIF use DET & RET)

Type	Name	RET	DET	Complexity (RET×DET)	FP Weight
ILF	User Access Events	2	14	Low (RET 2–5, DET 1–19)	7
EIF	External IdP	1	8	Low (RET=1, DET 1–19)	5

1.1. ELF- User Access Events: Your app maintains this history for traceability.

- RETs = 2 (Logical subgroup)

  · RET-1 Event core: login/logout/role change, result, time, session, IP, device, geo, role_at_time.
  · RET-2 MFA context: method and result (ties to the event).

- DETs= 14 event_id, user_id, action, result, timestamp, session_id, ip, device_id, geo, role_at_time, mfa_method, mfa_result, correlation_id, reason.

- RESULT: RET=2, DET=14

⇒ Low Complexity ⇒ 7 weight FP

1.2. EIF- External Identity Provider (IdP): Logical file owned outside your boundary

- RET: 1 (IdP account/token metadata as one subgroup)
- DET: 8 idp_user_id, idp_name, token, token_expiry, refresh_token, last_login_at, idp_status, federation_id.

- RESULT: RET=1, DET= 8

⇒ Low Complexity ⇒ weight 5 FP

2. Transaction Function Types (EI/EQ use DET & FTR)

Type	Name	FTR	DET	Complexity	FP Weight
EI	Login and Logout	2	8	Average (FTR=2, DET	4
EQ	Access Status	2	6	Average (FTR=2, DET	4

1.2. EI- Login & Logout: External input creates/updates session/auth state.

- FTRs read or maintained= 2. User Access Events (ILF), IdP (EIF)

- DETs entered or recognize= 8 username, credential/assertion, mfa_code, device_id, ip, user_agent, action_type, timestamp.

- RESULT: FTR=2, DET= 8

⇒ Average Complexity ⇒ weight 4FP

2.2. EQ- Access Status Query: A read-only inquiry returning current auth status

- FTRs referenced 2= User Access Events (ILF), IdP (EIF)

- DETs shown/returned = 6 user_id, session_id, current_role, mfa_state, last_activity_at, lockout_flag.

- RESULT: FTR=2, DET= 6

⇒ Low Complexity ⇒ weight 4 FP

Access Security subtotal (UFC) = 7 + 5 + 4 + 4 = 20 FP

D- System Availability:

1. Data Function Types (EIF uses DET & RET)

Type	Name	RET	DET	Complexity (RET×DET, Wk9 S17–18)	FP Weight
EIF	Monitoring Service API	2	10	Low (RET 2–5 & DET 1–19)	5

1.1. EIF- Monitoring Service API:

- RET = 2: two logical subgroups — metrics and incidents.
- DET = 10: service_id, metric_name, metric_value, time_window, region, status, incident_id, severity, started_at, ended_at.

⇒ Complexity: Low (RET 2–5, DET 1–19) ⇒ weight 5 FP.

2. Transaction Function Types (EO/EQ use DET & FTR)

Type	Name	FTR	DET	Complexity (FTR×DET, Wk9 S17–18)	FP Weight
EO	Availability Reports	1	12	Low (FTR 0–1 & DET 1–19)	4
EQ	Status Dashboards	1	10	Low (FTR 0–1 & DET 1–19)	3

1.2. EO- Availability Reports: Periodic, derived report (uptime/SLA over a window).

- FTR = 1: reads the Monitoring Service API (EIF) only.
- DET = 12: service_id, uptime_pct, downtime_minutes, incident_count, mttr, mtta, worst_region, best_region, peak_load, sla_target, sla_breaches, report_period.

⇒ Complexity: Low (FTR 0–1, DET 6–19) ⇒ weight 4 FP.

2.2. EQ- Status Dashboards: Immediate read-only inquiry

- FTR = 1: reads the Monitoring Service API (EIF).
- DET = 10: service_id, current_status, latency_p50, latency_p95, error_rate, active_incidents, affected_region, last_update, ticket_id, owner_team.

⇒ Complexity: Low (FTR 0–1, DET 6–19) ⇒ weight 3 FP.

System Availability subtotal (Unadjusted FP) = 5 + 4 + 3 = 12 FP

E-Compliance & Audit Readiness:

1. Data Function Types (ILF uses DET & RET)

Type	Name	RET	DET	Complexity (RET×DET)	FP Weight
ILF	Compliance Control / Audit Result Logs	3	22	Average (RET 2–5, DET 20–50)	10

1.1. ILF- Compliance Control / Audit Result Logs

- RET = 3: controls, tests, findings (3 logical subgroups).
- DET = the listed fields are user-recognizable/admin-visible.

⇒ AverageComplexity: RET 2–5 & DET 20–50 ⇒ weight 10 FP.

2. Transaction Function Types (EO/EQ use DET & FTR)

Type	Name	FTR	DET	Complexity (FTR×DET)	FP Weight
EO	Audit Reports	2	20	High (FTR 2–3, DET ≥20)	7
EQ	Control Queries	1	8	Low (FTR 0–1, DET 6–19)	3

-Compliance & Audit Readiness subtotal (UFC) = 10 + 7 + 3 = 20 FP.

1.2. EO- Audit Reports: Derived output summarizing compliance results.

- FTR = 2: reads Compliance/Audit ILF and Audit Log ILF.
- DET = 20: the report columns you listed.

⇒ High Complexity: EO with FTR 2–3 and DET >=20 ⇒ weight 7 FP.

2.2. EQ- Control Queries: Read-only inquiry of a specific control's status.

- FTR = 1: reads Compliance/Audit ILF.
- DET = 8: the fields returned on the inquiry.

⇒ Low Complexity: EQ with FTR 0–1 and DET 6–19 ⇒ weight 3 FP.

Step 3 — Compute Function Points (UFC, VAF, AFP)

A. Compute the Unadjusted Function Count (UFC):

To get UFC, we sum up the FP weights from Step 2 (for all ILFs, EIFs, EIs, EOs, EQs). These weights are based on their assigned complexity (Low, Average, High) and the FP weight Table.

Function Type	Count	Weight (per Type)	Total FP
ILF	3	10, 7, 10	27
EIF	2	5, 5	10
EI	2	6, 4	10
EO	4	7, 5, 4, 7	23
EQ	3	4, 3, 3	10
Total UFC			**80 FP**

UFC = 80 Function Points

B. Value Adjustment Factor (VAF):

VAF measures the extent to which the system is influenced by 14 General System Characteristics (GSCs).

Each characteristic is rated from **0 (no influence)** to **5 (strong influence).**

Then we apply the standard formula: VAF = 0.65 + (0.01 × TDI)

GSC	Description	Rating (0–5)
1	Data communications	3
2	Distributed functions	2
3	Performance requirements	4
4	Heavily used configuration	4
5	Transaction rate	2
6	Online data entry	5
7	End-user efficiency	4
8	Online update	4

9	Complex processing	3
10	Reusability	1
11	Installation ease	0
12	Operational ease	3
13	Multiple sites	2
14	Facilitate change	5
Total Degree of Influence (TDI)		**42**

$$VAF = 0.65 + (0.01 \times TDI)$$

$$VAF = 0.65 + (0.01 \times 48)$$

$$VAF = 1.13$$

C. Adjusted Function Points (AFP):

To calculate the adjusted function points, we use the standard formula:

$$AFP = UFC \times VAF$$

$$AFP = 80 \times 1.07$$

$$AFP = 85.6 \approx 86 \text{ FP}$$

Final Adjusted Function Point Count (AFP) = 86 FP